White-Box Testing Report

Moneypoly

Assignment 2

March 24, 2026

Contents

1 Scope **2**

2 1.1 Control Flow Graph (CFG) **2**

3 1.2 Code Quality Analysis (pylint) **2**

4 1.3 White Box Test Cases and Pytest Iterations **3**

4.1 Representative White-Box Cases 3

4.2 Logical Issues Found by Tests 3

5 Coverage Highlights **4**

1 Scope

This report documents white-box testing for the Moneypoly codebase, internal method checks, and targeted error-regression cases.

Primary test file:

- `whitebox/tests/test_monopoly_testing.py`

Code under test:

- `moneypoly.game`, `moneypoly.player`, `moneypoly.board`, `moneypoly.bank`
- `moneypoly.property`, `moneypoly.cards`, `moneypoly.dice`, `moneypoly.ui`
- Entry-point behavior in `main.py`

2 1.1 Control Flow Graph (CFG)

Figure 1 shows the whole-game control-flow graph used for white-box path and branch analysis.

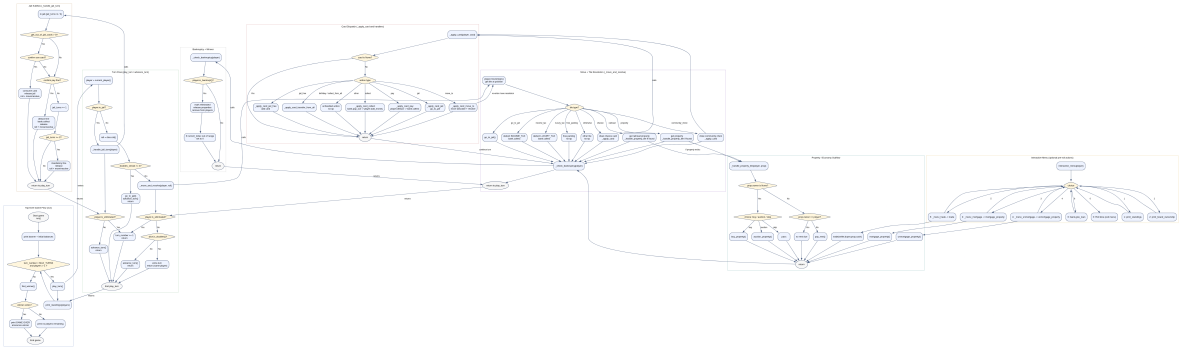

Figure 1: Whole-game CFG for Moneypoly (from `moneypoly.game`)

3 1.2 Code Quality Analysis (pylint)

The following pylint iterations are taken from the provided whitebox report data and recorded in sequence.

Pylint Iterations

1. **Iteration 1** – `board.py`: fixed missing module docstring and singleton-comparison style issue; re-run score reached `10.00/10`.
2. **Iteration 2** – `ui.py`: fixed missing module docstring and replaced bare `except` with `except ValueError`; final score `10.00/10`.
3. **Iteration 3** – `config.py`: added module docstring; final score `10.00/10`.
4. **Iteration 4** – `cards.py`: added module docstring and reformatted long card dictionary lines; final score `10.00/10`.

5. **Iteration 5** – `bank.py`: added module/class docstrings and removed unused `math` import; final score `10.00/10`.
6. **Iteration 6** – `dice.py`: added module docstring, removed unused import, initialized `doubles_streak` in `__init__`; final score `10.00/10`.
7. **Iteration 7** – `main.py`: added module and function docstrings; final score `10.00/10`.
8. **Iteration 8** – `player.py`: added module docstring, removed unused variable, added final newline, and adjusted attribute count pressure; final score `10.00/10`.
9. **Iteration 9** – `property.py`: added module/class docstrings, simplified `unmortgage`, reduced attribute pressure, and applied targeted constructor-argument lint handling; final score `10.00/10`.
10. **Iteration 10** – `game.py`: removed unused imports, simplified control paths, improved dispatch structure, and cleaned style warnings; final score `10.00/10`.

Environment Iteration Note

4 1.3 White Box Test Cases and Pytest Iterations

Test design targeted branch coverage, key state transitions, and edge conditions (zero/equality boundaries, invalid actions, and elimination/jail transitions).

4.1 Representative White-Box Cases

Representative passing categories covered by `test_monopoly_testing.py` include:

- property mortgage/unmortgage branches
- rent and ownership-state branches
- board tile routing and purchasable-state checks
- bank payout/loan guard paths
- jail handling and movement resolution branches
- trade/mortgage/unmortgage guard flows
- card action dispatch and unknown-action no-op
- turn-order safety after player elimination
- CLI input and main-entry error branches

4.2 Logical Issues Found by Tests

From the provided report data, the white-box suite exposed a broad set of logic issues across gameplay and accounting paths, including:

- missing purchasable railroad backing

- jail-fine cash inconsistency
- incorrect winner selection criterion
- salary-award boundary defects around Go
- trade cash-transfer and validation defects
- dice-range defect
- bank-loan and reserve-accounting defects
- net-worth computation defect
- card-resolution side-effect defects
- ownership and buy-flow validation defects

5 Coverage Highlights

Representative covered logic includes:

- Financial invariants (cash transfer, rent, loan, tax, salary)
- Rule enforcement (role, ownership, mortgage constraints)
- Defensive behavior for invalid operations and edge states
- UI helper branches and input parsing branches
- Main entry-point and error-handling paths